

Homework (Habanero)

- Q1.** A patient's temperature T (in $^{\circ}\text{F}$) is related to the time t (in hours) by the equation $T = 98.6 + 0.5t$. What is the rate of change of temperature with respect to time?
- (A) $0.25^{\circ}\text{F}/\text{hour}$
(B) $0.5^{\circ}\text{F}/\text{hour}$
(C) $0.8^{\circ}\text{F}/\text{hour}$
(D) $1.0^{\circ}\text{F}/\text{hour}$
(E) $1.2^{\circ}\text{F}/\text{hour}$
- Q2.** The graph below shows the relationship between the dosage of a medicine and its effectiveness. What is the slope of the line?
- (A) 2
(B) 1.5
(C) 1
(D) 0.5
(E) 0
- Q3.** A table of values for the number of bacteria N after t hours is given by: $t : 0, 1, 2, 3$ and $N : 100, 120, 140, 160$. What is the rate of change of the number of bacteria with respect to time?
- (A) 20 bacteria/hour
(B) 30 bacteria/hour
(C) 40 bacteria/hour
(D) 50 bacteria/hour
(E) 60 bacteria/hour
- Q4.** The equation $y = 2x - 3$ represents the relationship between the concentration of a chemical y and the amount of a substance x . What is the slope of this line?
- (A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q5.** A patient's heart rate is monitored over time, and the results are shown in the graph. If the slope of the line is -0.5 , what does this represent?
- (A) The patient's heart rate decreases by 0.5 beats per minute for every 1 minute that passes.
(B) The patient's heart rate increases by 0.5 beats per minute for every 1 minute that passes.
(C) The patient's heart rate remains constant.
(D) The patient's heart rate increases by 1 beat per minute for every 0.5 minutes that pass.
(E) The patient's heart rate decreases by 1 beat per minute for every 0.5 minutes that pass.
- Q6.** The slope of a line is -2 . What does this represent in the context of a patient's weight over time?
- (A) The patient gains 2 pounds per week.
(B) The patient loses 2 pounds per week.
(C) The patient's weight remains constant.
(D) The patient gains 1 pound per 2 weeks.
(E) The patient loses 1 pound per 2 weeks.
- Q7.** A line has an undefined slope. What does this represent in the context of a graph?
- (A) A vertical line.
(B) A horizontal line.
(C) A line with a slope of 0.
(D) A line with a slope of 1.
(E) A line with a slope of -1 .
- Q8.** The graph below shows the relationship between the amount of a medication and its effect on blood pressure. If the slope of the line is 3, what does this represent?
- (A) For every 1 unit of medication, blood pressure increases by 3 units.
(B) For every 3 units of medication, blood pressure increases by 1 unit.
(C) For every 1 unit of medication, blood pressure decreases by 3 units.
(D) For every 3 units of medication, blood pressure decreases by 1 unit.
(E) The relationship between medication and blood pressure is not linear.

Open Ended Questions (FRQ)

- Q9.** A patient's temperature T (in $^{\circ}\text{F}$) is related to the time t (in hours) by the equation $T = 98.6 + 0.5t$. Show that the rate of change of temperature with respect to time is constant, and calculate its value.
- Q10.** A table of values for the number of cells N after t hours is given by: $t : 0, 1, 2, 3$ and $N : 100, 150, 200, 250$. Calculate the rate of change of the number of cells with respect to time, and explain its significance in the context of cell growth.

Chef's Tips

- Chef's Tips: Make sure to check the units of the rate of change.
- Chef's Tips: Use the formula for slope to calculate the rate of change from a graph or table.
- Chef's Tips: Consider the context of the problem when interpreting the slope.